Pushing Medium (PM) — Consolidated Formula Sheet

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Core Assumptions

1. **Flat Minkowski background; curved effective metric.** The background geometry is fixed, flat Euclidean/Minkowski. No background GR curvature, no Einstein equations, no horizons, no TOV. The medium field $\phi = \ln n$ creates a *curved effective optical metric* $ds^2 = -(c^2/n^2)\,dt^2 + d\mathbf{r}^2$ for wave propagation — this is curvature of the medium, not of the background geometry.
2. **Scalar compression field.** Gravity arises from the scalar field ϕ with refractive index $n = e^{\phi}$. Sources: $\nabla^2\phi = -\frac{8\pi G}{c^2}\rho$.
3. **Density law.**
$$\rho = \rho_{\text{nuc}}\, e^{\phi}$$
4. **Force law for massive particles.**
$$\mathbf{a} = \tfrac{c^2}{2}\,\nabla\phi$$
(Half the photon coupling; recovers $-GM/r^2\,\hat{\mathbf{r}}$ exactly.)
5. **Light propagation.** Follows Fermat's principle in the optical metric:
$$ds^2_{\text{opt}} = -\frac{c^2}{n^2}\,dt^2 + (d\mathbf{r} - \mathbf{u}\,dt)^2$$
6. **Self-consistent EOS.**
$$P = \frac{c^2}{2}(\rho - \rho_{\text{nuc}})$$
7. **Stability cap.** The medium cannot be compressed beyond $\phi = 1$ ($n = e$, $\rho = e\,\rho_{\text{nuc}}$). This is the only strong-field limit in PM; it is not a compactness bound from GR.

Fields and Notation

Gravitational sector:

$$\phi(\mathbf{r},t) = \text{scalar compression field}, \quad \mathbf{u}(\mathbf{r},t) = \text{flow field}$$

Electromagnetic sector:

$$\psi(\mathbf{r},t) = \text{scalar EM field}, \quad \mathbf{w}(\mathbf{r},t) = \text{vector EM field}$$

Refractive index:

$$\boxed{n(\mathbf{r},t) = e^{\phi(\mathbf{r},t)}} \quad \Longleftrightarrow \quad \phi = \ln n$$

Static Gravity

Scalar field (Poisson):

$$\nabla^2\phi = -4\pi G_{\text{eff}}\,\rho_M$$

Point mass M:

$$\phi(r) = \frac{\mu_G M}{r}, \qquad n(r) = 1 + \frac{\mu_G M}{r} + \mathcal{O}(r^{-2})$$

Continuous distribution:

$$\phi(\mathbf{r}) = \mu_G \int \frac{\rho_M(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\,d^3r'$$

Vector flow field:

$$\nabla^2\mathbf{u} = -A_M\,\mathbf{J}_M$$

For a moving point mass with velocity $\mathbf{v}$:

$$\mathbf{u}(\mathbf{r}) = A_M M\,\frac{\mathbf{v}}{|\mathbf{r}-\mathbf{r}_M|}$$

Unified Source Structure

Scalar sources (gravity + EM + plasma):

$$S_\phi = 4\pi(G_{\text{eff}}\,\rho_M + K_E\,\rho_q + K_{\text{plasma}}\,n_e)$$

Vector sources (mass + charge currents):

$$\mathbf{S}_u = A_M\,\mathbf{J}_M + A_q\,\mathbf{J}_q$$

EM sources:

$$S_\psi = 4\pi K_E\,\rho_q, \qquad \mathbf{S}_w = B_q\,\mathbf{J}_q$$

Dynamic PM-Fluid Equations

Scalar field:

$$\frac{\partial\phi}{\partial t} + \nabla\cdot(\phi\,\mathbf{u}) = S_\phi + \frac{1}{c_\phi^2}\frac{\partial^2\phi}{\partial t^2}$$

Vector flow:

$$\frac{\partial\mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\nabla\phi + \mathbf{S}_u + \frac{1}{c_u^2}\frac{\partial^2\mathbf{u}}{\partial t^2}$$

PM Lagrangian Density

Gravitational sector:

$$\mathcal{L} = \frac{(\partial_t\phi)^2}{2c_\phi^2} - \frac{|\nabla\phi|^2}{2} + \frac{|\partial_t\mathbf{u}|^2}{2c_u^2} - \frac{|\nabla\mathbf{u}|^2}{2} + \phi\,\nabla\cdot(\phi\mathbf{u}) + \phi\,S_\phi + \mathbf{u}\cdot\mathbf{S}_u$$

Electromagnetic sector:

$$\mathcal{L}_{\text{EM}} = \frac{(\partial_t\psi)^2}{2c_\psi^2} - \frac{|\nabla\psi|^2}{2} + \frac{|\partial_t\mathbf{w}|^2}{2c_w^2} - \frac{|\nabla\mathbf{w}|^2}{2} + \psi\,S_\psi + \mathbf{w}\cdot\mathbf{S}_w$$

Structurally identical with $(\phi,\mathbf{u}) \to (\psi,\mathbf{w})$.

Hamiltonian Formulation

Canonical momenta:

$$\pi_\phi = \frac{\partial_t\phi}{c_\phi^2}, \qquad \pi_{u_i} = \frac{\partial_t u_i}{c_u^2}$$

Hamiltonian density:

$$\mathcal{H} = \frac{c_\phi^2\pi_\phi^2}{2} + \frac{|\nabla\phi|^2}{2} + \frac{c_u^2|\pi_u|^2}{2} + \frac{|\nabla\mathbf{u}|^2}{2} - \phi\,\nabla\cdot(\phi\mathbf{u})$$

Kinetic and gradient terms are positive-definite; the system is stable in the linear regime.

Hamilton's equations:

$$\partial_t \phi = c_\phi^2 \, \pi_\phi$$

$$\partial_t \pi_\phi = \nabla^2 \phi + \nabla \cdot (\phi \mathbf{u}) + \phi \, \nabla \cdot \mathbf{u}$$

$$\partial_t u_i = c_u^2 \, \pi_{u_i}$$

$$\partial_t \pi_{u_i} = \nabla^2 u_i + \partial_i (\phi^2)$$

Stress–Energy Tensor

Canonical form:

$$T^\mu{}_\nu = \sum_{\Psi \in \{\phi,\, u_i\}} \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Psi)} \, \partial_\nu \Psi - \delta^\mu{}_\nu \, \mathcal{L}$$

Energy density T^{00}:

$$\mathcal{E} = \frac{(\partial_t \phi)^2}{2c_\phi^2} + \frac{|\nabla \phi|^2}{2} + \frac{|\partial_t \mathbf{u}|^2}{2c_u^2} + \frac{|\nabla \mathbf{u}|^2}{2} - \phi \, \nabla \cdot (\phi \mathbf{u})$$

Momentum density T^{0i}:

$$T^{0i} = \frac{\partial_t \phi \, \partial_i \phi}{c_\phi^2} + \frac{\partial_t u_j \, \partial_i u_j}{c_u^2}$$

Stress tensor T^{ij}:

$$T^{ij} = -(\partial_i \phi)(\partial^j \phi) - (\partial_i u_k)(\partial^j u_k) - \delta^{ij} \, \mathcal{L}_{\mathrm{PM}}$$

Conservation:

$$\partial_\mu T^{\mu\nu} = 0 \quad \text{(no external sources)}$$

Electromagnetic Field Equations

Scalar EM (electric-like):

$$\frac{\partial \psi}{\partial t} + \nabla \cdot (\psi \, \mathbf{u}) = S_\psi + \frac{1}{c_\psi^2} \frac{\partial^2 \psi}{\partial t^2}$$

Static limit: $\nabla^2 \psi = -4\pi K_E \, \rho_q$

Vector EM (magnetic-like):

$$\frac{\partial \mathbf{w}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{w} = -\nabla \psi + \mathbf{S}_w + \frac{1}{c_w^2} \frac{\partial^2 \mathbf{w}}{\partial t^2}$$

Static limit: $\nabla^2 \mathbf{w} = -B_q \, \mathbf{J}_q$

Effective fields:

$$\mathbf{E}_{\mathrm{eff}} = -\nabla \psi - \partial_t \mathbf{w}, \qquad \mathbf{B}_{\mathrm{eff}} = \nabla \times \mathbf{w}$$

Maxwell-like equations (linear, no-flow limit):

$$\nabla \cdot \mathbf{E}_{\mathrm{eff}} = 4\pi K_E \, \rho_q, \qquad \nabla \cdot \mathbf{B}_{\mathrm{eff}} = 0$$

$$\nabla \times \mathbf{E}_{\mathrm{eff}} = -\partial_t \mathbf{B}_{\mathrm{eff}}, \quad \nabla \times \mathbf{B}_{\mathrm{eff}} = \frac{\partial_t \mathbf{E}_{\mathrm{eff}}}{c^2} + B_q \, \mathbf{J}_q$$

Ray and Particle Kinematics

Ray position:

$$\frac{d\mathbf{r}}{ds} = c \, \hat{\mathbf{k}} + k_{\mathrm{Fizeau}} \, \mathbf{u} \qquad (k_{\mathrm{Fizeau}} = 1)$$

Ray direction:

$$\frac{d\hat{\mathbf{k}}}{ds} = \nabla_\perp \phi + \mathcal{F}(\hat{\mathbf{k}}, \mathbf{u}, \nabla \mathbf{u})$$

Massive particle acceleration (exact, all speeds):

$$\boxed{\mathbf{a}_{\mathrm{PM}} = +\tfrac{c^2}{2} \, \nabla \phi}$$

Light sees both metric components (g_{tt} and g_{rr}) $\to$ factor c^2. Massive slow particles see only $g_{tt} \to$ factor $c^2/2$. With $\phi = \mu_G M/r$: $\mathbf{a} = -GM/r^2 \, \hat{\mathbf{r}}$ (Newtonian, exact).
Implementation: `core.massive_accel_medium`

Weak-field deflection angle:

$$\alpha_{\mathrm{PM}}(b) = \int_{-\infty}^{+\infty} \frac{-\mu_G M \, b}{n(r) \, r^3} \, dz \;\xrightarrow{n \to 1}\; \frac{2\mu_G M}{b} = \frac{4GM}{c^2 b}$$

Moving-lens correction:

$$\alpha_{\mathrm{moving}} = \alpha_{\mathrm{static}} \left(1 + \frac{v}{c}\right)$$

Effective Metric

Optical metric (exact):

$$ds_{\mathrm{opt}}^2 = -\frac{c^2}{n^2} \, dt^2 + (d\mathbf{r} - \mathbf{u} \, dt)^2$$

First-order expansion (small ϕ, small $\mathbf{u}$):

$$ds^2 = -(1 + 2\phi) c^2 dt^2 - 2 \, \mathbf{u} \cdot d\mathbf{r} \, dt + (1 - 2\phi) \, d\mathbf{r}^2$$

Metric components:

$$g_{tt} = -(1 + 2\phi), \qquad g_{rr} = (1 - 2\phi)$$

PPN parameters:

$$\gamma = 1, \qquad \beta = 1$$

Deformation Energy and Critical State

Landau–Ginzburg deformation energy:

$$U(\phi) = \varepsilon_0 \left(\phi^2 - \frac{\phi^3}{3}\right), \qquad \varepsilon_0 = \rho_{\mathrm{nuc}} \, c^2$$

$$U'(\phi) = \varepsilon_0 (2\phi - \phi^2), \qquad U''(\phi) = 2\varepsilon_0 (1 - \phi)$$

Stability criterion: $U''(\phi) > 0 \;\Leftrightarrow\; \phi < 1$

Critical point (onset of instability, $U'' = 0$):

$$\boxed{\phi_{\mathrm{crit}} = 1, \qquad n_{\mathrm{crit}} = e}$$

Density from field (exact):

$$\rho(\phi) = \rho_{\mathrm{nuc}} \, e^{\phi}$$

Self-consistent PM EOS (from $dP = \rho \frac{c^2}{2} d\phi$, $P \to 0$ at $\rho \to \rho_{\rm nuc}$):

$$\boxed{P = \tfrac{c^2}{2}(\rho - \rho_{\rm nuc})}$$

Properties: $P(\rho_{\rm nuc}) = 0$; sound speed $c_s = c/\sqrt{2}$.

Critical thermodynamic values:

$$\rho_{\rm crit} = e\,\rho_{\rm nuc} \approx 6.25 \times 10^{17}\ {\rm kg\,m^{-3}}\ \approx 2.7\,\rho_{\rm nuc}$$

$$P_{\rm crit} \approx 1.87 \times 10^{34}\ {\rm Pa}$$

Implementation: `src/pushing_medium/critical_state.py`

Diagnostic: Three-Way Disconnection of $U(\phi)$

The deformation energy $U(\phi)$ is currently used *only* as a stability condition. Three structural gaps exist.

Gap 1 — $U(\phi)$ is absent from the Lagrangian. The PM Lagrangian density (see §PM Lagrangian) contains gradient and kinetic terms $-\frac{1}{2}|\nabla\phi|^2 + \frac{(\partial_t\phi)^2}{2c_\phi^2}$ but no potential term $-V(\phi)$. Consequently the static equation of motion for ϕ is the linear Poisson equation

$$\nabla^2\phi = -\tfrac{8\pi G}{c^2}\rho_M,$$

with *no self-coupling from U*. The stability cap $\phi < 1$ is therefore an externally imposed condition, not a dynamical consequence of the Lagrangian. If U were included as $-V(\phi) = -U(\phi)/c_\phi^2$, the field equation would become

$$\nabla^2\phi - \frac{U'(\phi)}{c_\phi^2} = -\tfrac{8\pi G}{c^2}\rho_M \qquad \big[U'(\phi) = \varepsilon_0(2\phi - \phi^2)\big],$$

a nonlinear Poisson equation that would self-regulate compression and make $\phi_{\rm crit} = 1$ an emergent feature rather than a postulate.

Gap 2 — $U(\phi)$ and the EOS are thermodynamically inconsistent. If $U(\phi)$ were the fluid's internal energy density, the thermodynamic pressure would follow from $P_{\rm thermo} = \rho^2\partial(U/\rho)/\partial\rho$. Evaluating with $\rho = \rho_{\rm nuc}e^\phi$:

$$P_{\rm thermo} = \varepsilon_0\left(2\phi - 2\phi^2 + \frac{\phi^3}{3}\right).$$

At small ϕ this gives $P_{\rm thermo} \approx 2\varepsilon_0\phi$, whereas the stated EOS gives $P_{\rm EOS} = \frac{c^2}{2}(\rho - \rho_{\rm nuc}) \approx \frac{\varepsilon_0}{2}\phi$. The ratio $P_{\rm thermo}/P_{\rm EOS} \approx 4$ at lowest order. They are measuring fundamentally different things and are never reconciled in the current framework. $U(\phi)$ is thus *not* the thermodynamic free energy of the PM fluid — its physical interpretation is unspecified.

Gap 3 — Field gradient energy $\frac{1}{2}|\nabla\phi|^2$ does not gravitate. The Hamiltonian (see §Hamiltonian) contains $\frac{1}{2}|\nabla\phi|^2$ as positive-definite field energy, yet the Poisson equation sources ϕ from ρ_M alone. In the strong-field interior:

$$|\nabla\phi|^2 \sim \left(\frac{\mu_G m(r)}{r^2}\right)^2.$$

At the PM stability cap this omitted self-energy density is $\lesssim 1\%$ of $\rho_{\rm crit}c^2$, so it is negligible for stellar structure — but it is a formal inconsistency that would become important if $U(\phi)$ is modified to allow higher compactness.

Root cause. All three gaps trace to the same origin: $U(\phi)$ was introduced as a phenomenological stability condition (analogous to a Landau order parameter) without being derived from the PM Lagrangian. Because it feeds into neither the Poisson equation nor the hydrostatic equilibrium, the compact-star sector rests on a stability criterion that is decoupled from the dynamics. This is the most likely site of the strong-field aberration.

Minimal repair strategy. A self-consistent extension requires choosing *one* of:

(a) Add $-U(\phi)/c_\phi^2$ to the Lagrangian; re-derive the EOS from the modified equation of motion; accept that $P_{\rm EOS}$ changes.

(b) Keep the EOS and the Lagrangian as-is; replace $U(\phi)$ with a different stability argument whose critical point ϕ_c is a free parameter fit to compact-object data (effectively relaxing the hard cap).

(c) Interpret $U(\phi)$ as an *additional* elastic pressure of the medium, add it to the hydrostatic equilibrium ($P_{\rm total} = P_{\rm EOS} + U'(\phi)\times$(geometric factor)), and require thermodynamic self-consistency.

Numerical probe results (gap_probe_comparison.py). Running the PM compact-star ODE under each repair option yields:

- **Gap 2 — additional finding:** $P_{\rm thermo}(\phi)$ is *non-monotonic*. Its maximum occurs at

$$\frac{dP_{\rm thermo}}{d\phi} = \varepsilon_0(2 - 4\phi + \phi^2) = 0$$
$$\implies \phi_{\rm max} = 2 - \sqrt{2} \approx 0.586.$$

  By $\phi_{\rm crit} = 1$ the thermodynamic pressure has *decreased* to $P_{\rm thermo}(1) = \frac{1}{3}\varepsilon_0 < 0.80\,P_{\rm thermo}(\phi_{\rm max})$. Hence $U(\phi)$ cannot serve as a thermodynamic free energy anywhere in the range $[0.586, 1]$ — the derived "EOS" would have negative compressibility.

- **Gap 1 — Poisson correction magnitude (corrected formula):** The correct Gauss-theorem result for the $U'(\phi)$ self-coupling is

$$\delta\left(\tfrac{d\phi}{dr}\right) = \tfrac{\mu_G}{r^2}\int_0^r \tfrac{U'(\phi)\,r'^2}{c^2}\,dr'$$
$$\approx \mu_G\,\rho_{\rm nuc}(2\phi - \phi^2)\,\tfrac{r}{3}.$$

  (The naïve formula $(r/3)U'(\phi)/c^2$ omits $\mu_G = 2G/c^2$ and has unit ${\rm kg\,m^{-2}}$ instead of ${\rm m^{-1}}$, inflating the estimate by $\sim 10^{23}$.) The correctly normalised ratio at $\phi = \phi_{\rm crit}$ with interior density $\bar\rho \approx \rho_{\rm crit}$ is

$$\frac{\delta(d\phi/dr)}{(d\phi/dr)_{\rm std}} \approx \frac{\rho_{\rm nuc}}{4\pi\,\rho_{\rm crit}} = \frac{1}{4\pi e} \approx 2.9\%.$$

  The Gap 1 Poisson correction is a percent-level effect in the stellar interior — not negligible at precision level, but far smaller than the Gap 3 elastic pressure ($\sim 78\%$ of $P_{\rm EOS}$).

- **Option (b) — negative result:** Sweeping $\phi_{\rm crit}$ from 1 to 5 with the baseline linear EOS ($c_s = c/\sqrt{2}$) does *not* raise $M_{\rm max}$ above $30\,{\rm M_\odot}$ for any value in that range. The mass ceiling is determined by EOS stiffness, not by the stability cap. *Option (b) alone cannot close the BH mass gap.*

- **Option (c) — positive result:** Setting $P_{\rm total} = P_{\rm EOS} + U(\phi)$ at unchanged $\phi_{\rm crit} = 1$ raises $M_{\rm max}$ from

$\approx 14\,M_\odot$ (baseline) to $\approx 31\,M_\odot$ — comparable to the GW150914 lower component ($30.7^{+5.2}_{-3.0}\,M_\odot$). Mechanism: $U(\phi_{\rm crit}) = \frac{2}{3}\varepsilon_0 \approx 78\% \times P_{\rm EOS}(\rho_{\rm crit})$ substantially stiffens the effective EOS.

Gap 4 — $\rho_{\rm nuc}$ as the critical radius variable (from test_rho_nuc_constraint.py). After Option (c) fixes $M_{\rm max}$, the residual problem is the radius: $R_{1.4} \approx 14.0\,$km (ODE) vs. $\leq 13.3\,$km (GW170817, 90% CI). Further probing reveals the root cause:

The PM stellar surface condition is $P = 0 \Leftrightarrow \rho = \rho_{\rm nuc}$. A $1.4\,M_\odot$ star has mean density $\langle\rho\rangle \approx 0.11\,\rho_{\rm nuc}$, so the star is nearly a *uniform sphere at density* $\rho_{\rm nuc}$:

$$\boxed{R_{1.4} \approx \left(\frac{3\,M_{1.4}}{4\pi\,\rho_{\rm nuc}}\right)^{1/3} = 14.24\ \text{km}}$$

(ODE result: $14.0 \pm 0.1\,$km.) For this reason, $R_{1.4}$ is insensitive to both α (the U(ϕ) coefficient) and γ (the EOS sound-speed factor $c_s^2 = \gamma c^2/2$): both modify the EOS at densities well above $\rho_{\rm nuc}$, whereas the star's bulk is near $\rho_{\rm nuc}$. Numerical confirmation: $R_{1.4}$ varies $\leq 4\%$ over $\alpha \in [0, 2]$ and $\leq 8\%$ over $\gamma \in [0.3, 2]$.

The *only* parameter that directly controls $R_{1.4}$ is $\rho_{\rm nuc}$:

$$R_{1.4} \propto \rho_{\rm nuc}^{-1/3}, \qquad \frac{\partial \ln R_{1.4}}{\partial \ln \rho_{\rm nuc}} = -\tfrac{1}{3} \ \text{(confirmed to 0.2\%)}$$

Two-parameter repair and joint feasibility window. Defining $f = \rho_{\rm nuc}/\rho_{\rm nuc,0}$, the two observational constraints:

$$M_{\rm max} \geq 28\,M_\odot \quad \text{(requires } \alpha \sim 1\text{, i.e. Option c)}$$
$$R_{1.4} \leq 13.3\ \text{km} \quad \text{(requires } f \geq 1.18\text{)}$$

are *decoupled*: α controls $M_{\rm max}$ at high ϕ, f controls R via the surface condition. Both are simultaneously satisfiable:

f	$\rho_{\rm nuc}$ [kg m^{-3}]	$M_{\rm max}$	$R_{1.4}$	Status
1.00	2.30×10^{17}	$30.8\,M_\odot$	14.0 km	R too large
1.15	2.64×10^{17}	$28.7\,M_\odot$	13.3 km	marginal
1.18	2.71×10^{17}	$28.4\,M_\odot$	13.2 km	**BOTH OK**
1.20	2.76×10^{17}	$28.1\,M_\odot$	13.1 km	**BOTH OK**
1.22	2.81×10^{17}	$27.9\,M_\odot$	13.1 km	$M_{\rm max}$ low

(Here $\alpha = 1$ fixed; increasing α to ≈ 1.15 at $f = 1.18$ can simultaneously raise $M_{\rm max}$ back above $30\,M_\odot$.) The joint feasibility window is narrow ($\approx 3\%$ in $\rho_{\rm nuc}$) but real. A shift of $f = 1.18$ is physically reasonable: $\rho_{\rm nuc} = 2.71 \times 10^{17}$ is within the range of neutron-rich nuclear matter calculations, which typically predict saturation densities 10–30% above the symmetric-matter value of $2.3 \times 10^{17}\,$kg/m^3.

Summary of findings.

1. $M_{\rm max}$: controlled by $U(\phi)$ at high ϕ (Option c). Fixed.
2. $R_{1.4}$: controlled by $\rho_{\rm nuc}$ (the surface density anchor). Requires a $\sim 18\%$ upward revision of $\rho_{\rm nuc}$.
3. The two corrections are decoupled and jointly feasible.
4. $U(\phi)$ cannot serve as a thermodynamic free energy above $\phi = 0.586$.

Option (a) changes the most and preserves formal consistency. Options (b) and (c) are more conservative. Any repair must preserve the weak-field limit ($U \to 0$ as $\phi \to 0$) and the two-phase cosmological structure ($\phi_{\rm crit}$ couples to Ω_E).

Compact Star Structure (exact within PM)

PM uses a flat Minkowski background; the medium field ϕ creates a curved effective optical metric but there are no background-curvature GR corrections. Gauss's theorem on $\nabla^2\phi = -\mu_G \cdot 4\pi\rho$ (on the flat background) gives the structure ODE system exactly:

ODE system (interior, $r > 0$):

$$\frac{dm}{dr} = 4\pi r^2 \rho, \quad \frac{dP}{dr} = -\frac{G m(r)\,\rho}{r^2}, \quad \frac{d\phi}{dr} = -\frac{\mu_G\, m(r)}{r^2}$$

Boundary conditions (at centre $r \to 0$):

$$m(0) = 0, \quad P(0) = \tfrac{c^2}{2}(\rho_c - \rho_{\rm nuc}), \quad \phi(0) = \ln(\rho_c/\rho_{\rm nuc})$$

Surface: integrate until $P = 0$ ($\rho \to \rho_{\rm nuc}$, $\phi \to 0$).

Hard stability cap:

$$\rho_c < \rho_{\rm crit} = e\,\rho_{\rm nuc} \quad \Leftrightarrow \quad \phi_c < 1$$

(From $U''(\phi_{\rm crit}) = 0$; medium cannot be compressed past this.)

Self-consistency: ϕ from Poisson $= \ln(\rho/\rho_{\rm nuc})$ from EOS, verified algebraically. Implementation: `src/pushing_medium/stellar_structure.py`

Surface Gravitational Redshift

PM formula (from optical metric $g_{tt} = -c^2/n^2$, conserved frequency):

$$\boxed{z_{\rm PM} = e^{\mu_G M/R} - 1}$$

Interior field ϕ integrates to zero at the surface; the physically relevant quantity is the *exterior* Poisson solution $\phi_{\rm ext}(R) = \mu_G M/R$, giving a ratio of emitted to received frequency $\nu_\infty/\nu_s = e^{-\phi_{\rm ext}} = n(R)^{-1}$.

GR Schwarzschild formula (for comparison):

$$z_{\rm GR} = \frac{1}{\sqrt{1 - 2GM/c^2R}} - 1$$

Weak-field expansion (both theories):

$$z_{\rm PM} \approx \frac{2GM}{c^2R}, \qquad z_{\rm GR} \approx \frac{GM}{c^2R}$$

PM predicts $\approx 2\times$ larger surface redshift than GR for the same (M, R). This is a direct, falsifiable prediction testable with X-ray spectroscopy of neutron stars.

NICER spot-checks (central values):

Source	M [M$_\odot$]	R [km]	$z_{\rm PM}$	$z_{\rm GR}$	ratio
J0030+0451	1.34	12.71	0.365	0.205	1.78
J0740+6620	2.08	12.39	0.642	0.408	1.57

EXO 0748−676 (Cottam 2002): $z_{\rm obs} \approx 0.35$, consistent with PM at NICER masses but insufficient to distinguish the theories without an independent (M, R) measurement.

Status: structural prediction, not a gap. The factor-of-2 in $\mu_G = 2G/c^2$ is the *same* coefficient that produces

the correct $4GM/c^2b$ light deflection and the $2GM/c^3 \ln(\cdot)$ Shapiro delay. These cannot be adjusted without destroying PM's agreement with solar-system tests. The larger surface redshift is therefore a genuine, irreducible prediction of PM's optical-metric prescription — no choice of EOS parameters (f, α, γ, $\rho_{\rm nuc}$) can lower $z_{\rm PM}/z_{\rm GR}$ below ≈ 1.7 at neutron-star compactness. Numerical confirmation: $z_{\rm PM}/z_{\rm GR} = 1.78 \pm 0.02$ over all (f, α) variants tested.

Observational discriminant. A spectroscopic $z_{\rm surface}$ measurement that is *independent* of (M, R) would sharply distinguish the theories. If any well-characterised neutron star (e.g. from NICER timing + hotspot modelling) yields $z \lesssim 0.25$, PM's optical-metric formula is falsified. If $z \gtrsim 0.35$ is confirmed, it favours PM over GR.

Open question. The formula $1 + z = e^{\phi_{\rm ext}}$ is derived from the *static* ratio of g_{tt} at emission and reception in the optical metric. In a physical dispersive medium, however, the photon frequency also evolves as it propagates through a spatially varying refractive index (analogous to adiabatic wavelength stretching in fibre optics). Whether the medium wave-equation derivation reproduces the static g_{tt} ratio exactly, or introduces a correction factor, is not yet settled. If the wave-equation result differs, the $\approx 2\times$ discrepancy could be reduced without touching μ_G.

Implementation: `stellar_structure.pm_surface_redshift`, `classical.gr_surface_redshift`, `scripts/surface_redshift_track.py`, `tests/test_z_surface_variants.py`

GW Ringdown: PM vs GR

GR Schwarzschild QNM (l=2, n=0 fundamental; Leaver 1985):

$$f_{\rm QNM} = \frac{0.37367}{2\pi}\frac{c^3}{GM} \approx 12.1\,{\rm kHz} \times \frac{M_\odot}{M}$$

$$\tau_{\rm QNM} = \frac{GM}{2 \times 0.08896\, c^3} \approx 27.7\,\mu{\rm s} \times \frac{M}{M_\odot}, \quad Q_{\rm GR} \approx 2.10$$

(critically damped: energy absorbed by the horizon.)

PM: no horizon → no QNMs. The PM photon sphere lies at:

$$r_{\rm ps}^{\rm PM} = \mu_G M = \frac{2GM}{c^2} \quad \left(\tfrac{2}{3} \text{ of GR value } 3GM/c^2\right)$$

For realistic PM stars $R_\star > r_{\rm ps}^{\rm PM}$, so there are no exterior trapped photon orbits and no horizon QNMs.

PM surface (breathing) mode:

$$f_{\rm surf} = \frac{c_s}{2R_\star} = \frac{c}{2\sqrt{2}\,R_\star} \approx \frac{9.6\,{\rm kHz}}{R_\star/11\,{\rm km}}$$

$$\tau_{\rm surf} = \frac{Q}{\pi f_{\rm surf}}$$

($Q_{\rm PM} \gg Q_{\rm GR} \approx 2.1$; no horizon energy sink.)

PM surface echo delay:

$$\tau_{\rm echo} = \frac{2R_\star}{c} \approx 73\,\mu{\rm s} \times \frac{R_\star}{11\,{\rm km}}$$

PM maximum compact-star mass. Newtonian structure (no TOV); mass grows monotonically with ρ_c up to the stability cap:

$$\rho_c \le \rho_{\rm crit} = e\,\rho_{\rm nuc} \quad \Longrightarrow \quad M_{\rm max}^{\rm PM} \approx 13\text{–}14\,M_\odot$$

LIGO falsification summary:

- GW150914 ($29 + 36\,M_\odot$ progenitors): both exceed $M_{\rm max}^{\rm PM}$; PM predicts no compact progenitors.
- GW170817 ($1.46 + 1.27\,M_\odot$): within PM range, but PM predicts $R \approx 15$ km vs NICER ≈ 12 km (+30% offset).
- At $M \approx 2\,M_\odot$: $f_{\rm surf} \approx 8$ kHz $> f_{\rm QNM} \approx 6$ kHz (factor-of-1.3 difference; accessible to Einstein Telescope).

Impl.: `core.pm_surface_echo_delay`, `classical.gr_qnm_frequency`, `scripts/gw_ringdown_comparison.py`

Black-Hole Shadow / EHT: PM vs GR

GR Schwarzschild BH shadow (photon-sphere capture impact parameter):

$$b_{\rm GR}^{\rm BH} = 3\sqrt{3}\,\frac{GM}{c^2} \approx 2.598\,R_s$$

arising from the photon sphere at $r_{\rm ps} = 3GM/c^2$.

GR compact star shadow (surface-grazing Schwarzschild ray):

$$b_{\rm GR}^\star(R_\star) = \frac{R_\star}{\sqrt{1 - R_s/R_\star}}$$

equals $b_{\rm GR}^{\rm BH}$ when $R_\star = r_{\rm ps} = 3GM/c^2$.

PM compact star shadow. In the PM exterior $n(r) = \exp(\mu_G M/r)$, Bouguer's theorem gives $b = n(r)\,r\,\sin\theta =$ const. The critical impact parameter for a ray grazing the stellar surface is:

$$\boxed{b_{\rm PM}^\star(R_\star) = e^{2GM/(c^2 R_\star)}\,R_\star}$$

PM has no photon sphere for $R_\star \ge R_s$, proven analytically: $F'(r) = \frac{d}{dr}[n(r)\,r] > 0$ for all $r \ge R_s$.

Key comparison at maximum compactness ($R_\star \to R_s$):

$$b_{\rm PM}^{\rm min} = e\,R_s \approx 2.718\,R_s \;>\; b_{\rm GR}^{\rm BH} \approx 2.598\,R_s$$

$$\frac{b_{\rm PM}^{\rm min}}{b_{\rm GR}^{\rm BH}} = \frac{2e}{3\sqrt{3}} \approx 1.046$$

PM always predicts shadows $\ge 4.6\%$ *larger than a GR BH of the same mass.* For typical PM neutron-star compactness ($k = R_\star/R_s \approx 3$) the excess is $\approx$60%.

	b/R_s	GR BH?	Shadow prediction
GR BH	$3\sqrt{3}/2 \approx 2.598$	yes	$\theta_{\rm GR}$
PM max-compact ($k = 1$)	$e \approx 2.718$	no	+4.6% over GR BH
PM typical NS ($k = 3$)	$e^{1/3} \times 3 \approx 4.187$	no	+61% over GR BH

EHT observations and PM double falsification:

Source	$M\,[M_\odot]$	D	$\theta_{\rm obs}$ [μas]	$\theta_{\rm GR}$	$M/M_{\rm max}^{\rm PM}$
M87*	6.5×10^9	16.8 Mpc	42 ± 3	39.7	$\sim 4.6 \times 10^8$
Sgr A*	4.15×10^6	8.178 kpc	51.8 ± 2.3	52.1	$\sim 3.0 \times 10^5$

1. **Mass-gap falsification:** EHT source masses (10^6–$10^9\,M_\odot$) exceed the PM maximum compact-object mass ($\approx 14\,M_\odot$) by factors of 10^5–10^8. PM cannot form these objects.
2. **Shadow-shape falsification:** If PM compact objects *could* exist at these masses, even the most compact PM case would predict a shadow 4.6% larger than GR. For any physically realistic compactness the excess is 10–160%. EHT observations are consistent with GR to $\lesssim 5\%$.

Implementation: `classical.gr_shadow_radius`, `core.pm_shadow_radius`, `tests/test_eht_shadow.py` (23 tests), `scripts/eht_shadow_comparison.py`

Photon-Sphere Location Across the M–R Curve

The PM photon sphere sits at $r_{\rm ps} = 2GM/c^2$ (the Schwarzschild radius, 2/3 of GR's $3GM/c^2$). Whether $r_{\rm ps}$ lies inside or outside the stellar surface depends on the compactness $C = GM/(c^2R)$:

Regime	Compactness	$r_{\rm ps}$ vs $R_\star$	Consequence
$C < 0.5$ (low mass)	$C \lesssim 0.49$	inside	no photon ring
$C > 0.5$ (near $M_{\rm max}$)	$C \lesssim 0.74$	outside	photon ring exists

Numerical results from the PM M–R curve:

Mass	$R_\star$	C	$r_{\rm ps}$	$r_{\rm ps}$ location
$1.4\,M_\odot$ (typical NS)	$\approx 14\,$km	0.148	$\approx 4.1\,$km	inside star
$3\,M_\odot$	$\approx 18\,$km	0.24	$\approx 8.9\,$km	inside star
$\sim 8\,M_\odot$ (transition)	$\approx 24\,$km	0.50	$\approx 24\,$km	at surface
$13.4\,M_\odot$ (max)	$\approx 26.6\,$km	0.74	$\approx 39.5\,$km	outside ($1.49 \times R_\star$)

Why can PM exceed $C = 0.5$? GR imposes the Buchdahl limit ($C < 4/9 \approx 0.444$ for uniform-density stars) and the horizon condition ($C = 0.5$ marks the Schwarzschild radius). PM uses flat-space Newtonian structure — there is no Buchdahl limit and no event-horizon formation regardless of C. High-mass PM stars are simply ultracompact, horizonless, optically opaque objects.

Key findings:

- All PM stars satisfy $C < 1.0$ (no horizon-scale compactness).
- The Buchdahl limit $C < 4/9$ is *exceeded* by PM near $M_{\rm max}$.
- Low-mass NS-like models ($M \lesssim 8\,M_\odot$) have the photon sphere safely interior — consistent with the original claim in the QNM section.
- High-mass near-$M_{\rm max}$ models have an *exterior* photon ring — a PM-specific structural prediction.

Status: structural prediction; not directly constrained by current observations. Future light-ring sensitive observations (GW ringdown, EHT-like instruments pointing at PM compact objects) could probe this.

Implementation: `core.pm_photon_sphere_radius`; `pushing_medium.stellar_structure.compute_mr_curve`; `tests/test_photon_sphere_guarantee.py` (18 tests).

GW Inspiral / Chirp Waveform: PM vs GR

0PN Identity Theorem

At leading (quadrupole) order, PM and GR produce *identical* inspiralling waveforms. The two ingredients are:

1. Quadrupole radiation power $P_{\rm GW} = \frac{32}{5}\frac{G^4M_1^2M_2^2(M_1+M_2)}{c^5\,a^5}$ (same formula; Hulse–Taylor validates to $< 0.2\%$),
2. Newtonian orbital energy $E = -\frac{GM}{2a}$ (same).

Because $\dot f \propto M_c^{5/3} f^{11/3}$ depends only on these, the *chirp mass* measured by LIGO is indistinguishable between PM and GR at this order.

Key Formulas (0PN, circular orbit)

Chirp mass (directly measured from df/dt):

$$\boxed{M_c = \frac{(M_1M_2)^{3/5}}{(M_1+M_2)^{1/5}}}$$

$$\frac{df}{dt} = \frac{96}{5}\,\pi^{8/3}\left(\frac{GM_c}{c^3}\right)^{5/3} f^{11/3} \qquad \text{(frequency sweep)} \tag{1}$$

$$t_{\rm coal}(f) = \frac{5}{256}\left(\frac{c^3}{GM_c}\right)^{5/3}(\pi f)^{-8/3} \qquad \text{(time to merger)} \tag{2}$$

$$h_c = \frac{4}{D}\,\frac{GM_c}{c^2}\left(\frac{\pi GM_c f}{c^3}\right)^{2/3} \qquad \text{(peak strain, face-on)} \tag{3}$$

The **analytic 0PN chirp** ($\tau = t_{\rm coal} - t$):

$$f(\tau) = \frac{5^{3/8}}{8\pi}\left(\frac{c^3}{GM_c}\right)^{5/8}\tau^{-3/8} \tag{4}$$

$$\Phi(\tau) = \Phi_{\rm ref} - 2\left(\frac{\tau}{\tau_c}\right)^{5/8}, \quad \tau_c = \frac{5\,GM_c}{c^3} \tag{5}$$

GWTC-3 Mass Gap: the Dominant PM Constraint

Differences between PM and GR arise only at 1.5PN+ (tail terms, near-zone corrections). The primary falsification is **object formation**: PM sets a compact-object mass ceiling $M_{\rm max} \approx 13$–$14\,M_\odot$, yet the majority of LIGO events have components far above this.

Event	$M_1/M_\odot$	$M_2/M_\odot$	$M_c/M_\odot$	PM accessible?
GW150914	35.6	30.6	28.7	**No — gap**
GW151226	13.7	7.7	8.9	Marginal/No
GW170814	30.5	25.3	24.2	**No — gap**
GW170817	1.46	1.27	1.19	Yes (BNS)
GW190814	22.2	2.6	6.0	**No — gap**
GW190521	85.0	66.0	65.1	**No — gap**

Quantified gap (ODE result): $M_{\rm max}^{\rm baseline} \approx 13.4\,M_\odot$; nearest GWTC heavy object is GW190814 M1 = $22.2\,M_\odot$. Gap = $8.8\,M_\odot$ (factor 1.66). 4 of the 6 events surveyed have both components above PM $M_{\rm max}$; only GW170817 (BNS) is fully accessible.

Implementation: `tests/test_mass_gap.py` (22 tests, 4 classes: `TestBaselineMMax`, `TestMassGapGWTC`, `TestBNSAccessible`, `TestMassGapCharacterisation`).

Option-A: Corrected Poisson Equation — Numerical Experiment

Motivation. The Gap-1 diagnostic (Deformation Energy section) identifies that $U(\varphi)$ is absent from the PM Lagrangian. Promoting it into the field equation adds a self-coupling source term. We parameterise the correction by α:

$$\nabla^2\varphi \;=\; -\frac{8\pi G}{c^2}\big[\rho \;-\; \alpha\,\rho_U\big], \qquad \rho_U \;=\; \rho_{\rm nuc}(2\varphi - \varphi^2)$$

Via Gauss's theorem inside a star of effective mass $m_U(r) = 4\pi \int_0^r \rho_U r'^2\, dr'$:

$$\frac{d\varphi}{dr} = -\frac{\mu_G\big(m(r) - \alpha\, m_U(r)\big)}{r^2}$$

Two physically motivated directions:

- $\alpha = +1$ (**Gap-1**, formula-sheet direction): U' reduces the gravitating source, weakening gravity at high compression.
- $\alpha = -1$ (**Gap-3**): U' adds to gravity, field energy itself gravitates.

Numerical results (50 central densities, each scenario):

Scenario	$M_{\max}$ [$M_\odot$]	$R_{1.4}$ [km]	$C_{\max}$	$z_s^{\max}$	$r_{\rm ps}/R_{\max}$
Baseline ($\alpha = 0$)	13.40	13.92	0.743	3.42	1.486
Gap-1 ($\alpha = +1$)	30.28	13.94	1.280	11.94	2.560
Gap-3 ($\alpha = -1$)	7.94	13.89	0.524	1.85	1.048
GW170817 bound: $R_{1.4} \le 13.3$ km					

$\varphi(r)$ profile at $\rho_c = 1.4\,\rho_{\rm nuc}$:

r/R	$\varphi_{\rm base}$	$\varphi_{\rm Gap1}$	$\varphi_{\rm Gap3}$
0.0	0.336	0.336	0.336
0.5	0.246	0.254	0.241
1.0	0.000	0.000	0.000

Option-A: Key Findings

1. **$M_{\max}$ is highly sensitive to the correction.** Gap-1 doubles $M_{\max}$: $13.4 \to 30.3\,M_\odot$. This crosses the GW190814 primary ($22.2\,M_\odot$) and is comparable to GW150914 components (30–36 $M_\odot$). The GWTC mass gap is dramatically narrowed.
2. **$R_{1.4}$ is insensitive to the correction.** All three scenarios give $R_{1.4} \approx 13.9$ km, all still exceeding the GW170817 bound of 13.3 km by ~ 0.6 km. The radius problem is *not* fixed by the Poisson correction.
3. **The two problems are decoupled.** The Poisson correction activates primarily at high compression (near $M_{\max}$ where $\rho \gg \rho_{\rm nuc}$). At $1.4\,M_\odot$, $\varphi_c \approx 0.34$ and the φ^2 correction term is subdominant, so $R_{1.4}$ is unaffected. The radius problem requires a *separate* EOS adjustment (Option-C: $\rho_{\rm nuc}$ rescaling).
4. **Gap-1 at $M_{\max}$ produces extreme compact objects.** $C = 1.28$, $z_s = 11.94$ — photon sphere at $2.56\,R$, outside the star. No GR horizon exists (flat PM background), but such objects would have extreme surface redshifts unlike any observed neutron star.
5. **EM cross-check is unaffected.** At low mass ($1.4\,M_\odot$), $\varphi(r)$ changes by $< 10\%$ at $r/R = 0.7$. EHT shadow, photon-sphere, and surface-redshift predictions for ordinary-density stars are effectively unchanged.

Combined Option-A + Option-C strategy: Since the two problems decouple, both could be fixed simultaneously: Gap-1 correction addresses $M_{\max}$; EOS rescaling (Option-C) addresses $R_{1.4}$. These are independent degrees of freedom.

Implementation: `src/pushing_medium/stellar_structure.py` (`solve_pm_star_option_a`, `compute_mr_curve_option_a`), `scripts/option_a_corrected_poisson.py`, `tests/test_option_a_corrected_poisson.py` (21 tests, 4 classes).

GW170817 tidal deformability (computed): Computed via the Clairaut–Radau equation for the Newtonian Love number k_2 (Poisson & Will *Gravity* eq. 14.34):

$$\frac{dy}{dr} = \frac{-y^2 - 5y + 6\nu(1+y)}{r}, \quad \nu(r) = \frac{\rho(r)}{\bar\rho(r)}, \quad y(0) = 2$$

$$k_2 = \frac{y_R - 2}{2(2y_R + 3)}, \qquad \Lambda = \frac{2}{3}\frac{k_2}{C^5}, \quad C = \frac{GM}{c^2 R}$$

Config	R [km]	C	y_R	k_2	Λ
Baseline f=1.00, α=0	13.94	0.1479	2.798	0.0464	437
Option-c f=1.00, α=1	13.94	0.1479	2.798	0.0464	437
JointFeas f=1.18, α=1	13.18	0.1567	2.786	0.0459	324
Tighter f=1.20, α=1	13.10	0.1574	2.785	0.0458	316
GW170817: $\tilde\Lambda < 800$ (90% CI), < 580 (50% CI)					
GR reference: SLy ≈ 250, DD2 ≈ 660					

Key finding: PM tidal deformability is *within* GW170817 bounds for all configurations. The joint feasibility configurations ($f = 1.18$–1.20) give $\Lambda \approx 315$–325, comparable to the APR4 GR EOS ($\Lambda \approx 320$) and well below both the 90% and 50% CI upper bounds. The earlier estimate $\Lambda_{\rm PM} \approx 3.7\,\Lambda_{\rm GR}$ was for the wrong $f = 1$ baseline and used a R^5 scaling that does not account for the Love number k_2 reduction in a near-uniform-density PM star. The nearly uniform PM density profile ($\nu(r) \approx 1$ throughout) gives $y_R \approx 2.8$, $k_2 \approx 0.046$, which partially cancels the larger radius effect vs. GR EOS with $\Lambda \approx 300$–660.

Implementation: `tests/test_tidal_deformability.py` (14 tests).

Implementation: `core.pm_chirp_mass`, `pm_gw_frequency_deriv`, `pm_time_to_coalescence`, `pm_gw_strain_amplitude`, `pm_gw_chirp_waveform`; `classical.gr_chirp_mass`, `gr_time_to_coalescence`, `gr_gw_strain_amplitude`; `tests/test_gw_inspiral.py` (36 tests), `scripts/gw_inspiral_comparison.py`

GW Polarisation: Scalar Breathing Mode

Question: Does the PM scalar field $n(\mathbf{r}, t)$ produce a breathing-mode (scalar, l=0) GW polarisation during binary inspiral, in violation of the LIGO O3 bound?

Conservation-law proof (no leading-order breathing radiation):

1. **Monopole:** total mass $M_{\rm tot} = \text{const} \Rightarrow$ no monopole scalar radiation.
2. **Dipole:** centre-of-mass fixed for an isolated binary $\Rightarrow$ no dipole scalar radiation.
3. **Scalar quadrupole:** for a *circular* orbit the scalar trace of the mass quadrupole

$$I_{\rm trace} = \sum_i m_i\, |\mathbf{r}_i|^2 = \mu a^2 = \text{const}$$

is *constant* (unlike the STF tensor $\tilde I_{ij}$ which oscillates at 2ω). Hence $\ddot I_{\rm trace} = 0$ exactly, and there is **zero leading-order scalar quadrupole radiation**.

First non-zero breathing contribution (from orbital decay):

$$\ddot{I}_{\rm trace} = -\frac{4\mu\beta^2}{a^6}, \qquad \beta = \frac{64}{5}\frac{G^3 M_1 M_2 M_{\rm tot}}{c^5}$$

Ratio of breathing strain to tensor h_+:

$$\frac{|h_b|}{|h_+|} \approx \frac{\beta^2}{\omega^2 a^8} \sim \left(\frac{v}{c}\right)^{10} \times \text{(mass factor)}$$

Numerical result (GW170817 parameters, $f_{\rm GW}$ = 100 Hz):

Quantity	PM prediction	Bound
$\lvert h_b\rvert/\lvert h_+\rvert$	$\approx 10^{-7}$	(unobservable)
$\alpha_{\rm NT} = P_s/P_{\rm tot}$	$\approx 10^{-14}$	< 0.07 (O3)
Margin above O3	$\sim 10^{13}$	—

Key finding: PM predicts effectively zero breathing-mode GW radiation for quasi-circular binary inspiral. The non-tensor power fraction $\alpha_{\rm NT} \approx 10^{-14}$ is 13 orders of magnitude below the LIGO O3 upper limit of 0.07 (Abbott et al. 2021, PRD 103, 122002).

PM passes the GW polarisation test.

Implementation: `core.pm_scalar_quadrupole_trace`, `pm_breathing_strain_ratio`, `pm_nontensor_power_fraction`; `tests/test_gw_polarisation.py` (20 tests).

Perihelion Precession: PM vs GR

PM force law is exactly Newtonian:

$$\mathbf{a} = \tfrac{c^2}{2}\nabla\phi = -\frac{GM}{r^2}\hat{r}$$
$$\implies \Delta\omega_{\rm force} = 0 \text{ (closed Keplerian ellipse)}$$

1PN metric correction (optical metric $g_{tt} = -c^2/n^2$, $g_{rr} = 1 - 2\phi$; PM PPN parameters $\beta = \gamma = 1$, identical to GR):

$$\boxed{\Delta\omega = \frac{6\pi GM}{c^2\, a\,(1-e^2)}}$$

(rad per orbit; same formula as GR at 1PN.) The precession is a *metric / coordinate* effect, not a direct consequence of the PM force law.

Solar-system verification (1PN formula vs. observed GR residuals):

Body	a [AU]	e	PM 1PN [″/cy]	Obs. [″/cy]
Mercury	0.3871	0.2056	42.99	42.98
Venus	0.7233	0.0068	8.63	8.62
Earth	1.0000	0.0167	3.84	3.84
Mars	1.5237	0.0934	1.35	1.35

Agreement $< 0.04\%$ for all planets.

Binary pulsar PSR B1913+16 ($M_{\rm tot} = 2.83\,M_\odot$, $P_b = 7.75\,$h, $e = 0.617$):

$$\dot{\omega}^{\rm 1PN}_{\rm PM} = 4.228\,\text{deg/yr}$$

Observed: 4.22663 ± 0.00002 deg/yr (+0.04%).

Key distinction: GR precession is a real closure failure of the purely flat geodesic. In PM the flat-space trajectory is exactly Keplerian; the precession is an artifact of the optical-metric coordinate transformation (how light signals define angles). At 1PN the two are observationally identical. At 2PN they differ at the $\sim 10^{-8}$ ″/cy level (unmeasurable in the solar system).

Numerical verification: Mercury 1PN orbit integrator (DOP853, $n = 10$ orbits) agrees with the analytic formula to $< 0.001\%$ and confirms pure-Newtonian run gives exactly zero precession.

Implementation: `core.pm_integrate_orbit`, `core.pm_precession_arcsec_per_century`, `scripts/perihelion_precession_comparison.py`

Cosmological Structure Growth: PM vs ΛCDM

PM Refractive-Redshift Background

In the PM framework, cosmological redshift arises from a monotonically decreasing refractive index $n(t)$: the photon wavelength stretches as $1+z = n(t_{\rm emit})/n(t_{\rm obs})$. This yields an effective Hubble rate

$$H_{\rm PM}(z;\beta) = H_0\,(1+z)^\beta, \qquad \beta_{\rm calib} = 0.8.$$

With $\beta = 0.8$ the luminosity distance D_L matches ΛCDM to within $\sim 3\%$ over $z = 0.1$–1.5:

z	$D_L^{\Lambda\rm CDM}$ [Mpc]	$D_L^{\rm PM}$ [Mpc]	Diff [%]
0.1	478	471	−1.4
0.5	2927	2818	−3.7
1.0	6803	6614	−2.8
1.5	11205	11182	−0.2

Despite matching D_L, the *expansion history* differs: at $z = 2$, $H_{\rm PM}(2) = 2.41\,H_0$ vs. $H_{\Lambda\rm CDM}(2) = 3.03\,H_0$. Less Hubble friction means PM growth is *faster* than ΛCDM at high redshift.

PM Structure-Growth Scenarios

Scenario 1 — PM-Static (density perturbation in $H_{\rm PM}$ background with no friction term from the drag force):

$$\ddot{\delta} = 4\pi G\rho_m\,\delta \implies \delta(t) \propto e^{\Gamma t},$$
$$\Gamma = \sqrt{\tfrac{3}{2}}\,H_0\sqrt{\Omega_m} \approx 0.687\,H_0$$

The growth rate observable $f_{\rm static}(z) = \Gamma/H_{\rm PM}(z)$ *increases with redshift* (opposite to ΛCDM), grossly overpredicting cluster abundances.

Scenario 2 — PM-Drag (index decay provides Hubble-like friction):

$$D'' + \frac{3-\beta}{a}\,D' - \frac{3\Omega_{\rm eff}}{2\,a^{5-2\beta}}\,D = 0$$

where primes denote d/da and $E(a) = a^{-\beta}$. The characteristic power-law growing mode $D \propto a^p$ satisfies

$$p^2 + (2-\beta)\,p - \tfrac{3}{2}\Omega_{\rm eff} = 0$$
$$p = \tfrac{1}{2}\left[(\beta-2) + \sqrt{(2-\beta)^2 + 6\Omega_{\rm eff}}\right].$$

For $\beta = \frac{3}{2}$ (ΛCDM matter-domination), $p = 1$ exactly.

Falsification — runaway growth at $\Omega_{\rm eff} = 1$: For $\beta = 0.8$, the gravity term $\propto a^{-3.4}$ grows much faster than the friction term $\propto a^{-2}$ at early times. There is no valid power-law matter-dominated era; the ODE

solver (DOP853) fails with step-size exhaustion. PM with $\Omega_{\text{eff}} = 1$ and $\beta = 0.8$ predicts catastrophic exponential over-growth — *excluded*.

$f\sigma_8$ Comparison with RSD Data

The growth rate observable is $f\sigma_8(z) = f(z)\,\sigma_{8,0}\,D(z)/D(0)$.

Model	χ^2 (N=14)	χ^2/N
ΛCDM (Ω_m=0.315, σ_8=0.811)	12.45	0.889
PM-Static β=0.8	23.02	1.644
PM-Drag β=0.8, Ω_{eff}=0.315	10.44	0.746
PM-Drag β=0.8, Ω_{eff}=1.0	ODE fails (runaway)	

Data: 14 published RSD $f\sigma_8$ measurements, $z = 0.02$–1.40 (6dFGS, SDSS MGS, WiggleZ, BOSS LOWZ/CMASS, FastSound, etc.).

Key conclusions:

1. PM-Drag with $\Omega_{\text{eff}} = 0.315$ achieves marginally *lower* χ^2/N than ΛCDM on current data, but only because it effectively uses Ω_{eff} as a free parameter separately from the D_L calibration. There is *no* parameter-free PM prediction.
2. PM-Static is excluded at $\chi^2/N \approx 1.6$ (wrong slope).
3. PM with $\Omega_{\text{eff}} = 1$ (no dark energy) is excluded by ODE failure: it cannot reproduce a matter-dominated era.
4. A joint fit $[\beta, \Omega_{\text{eff}}, \sigma_{8,0}]$ constrained simultaneously by D_L supernovae + $f\sigma_8$ RSD surveys would tightly constrain or rule out the refractive model.

Implementation: `src/pushing_medium/cosmology.py`, `src/general_relativity/cosmology.py`, `scripts/fsigma8_comparison.py`, `tests/test_cosmology_growth.py` (35 tests).

PM Two-Phase Cosmology: $\beta = 0.8$ as Dark Energy in Disguise

The Phase Transition Connection

PM has a natural phase transition at $\varphi_{\text{crit}} = 1$ (compact objects) where dense matter transitions to a uniform "energy phase" whose density does *not* dilute with expansion. Cosmologically, this provides a dark-energy analogue with two components:

Phase	Density scaling	Behaviour
Matter phase	$\rho_m \propto a^{-3}$	Clusters; growth
Energy phase	$\rho_E \propto a^{-3(1+w_E)}$	No clustering; $w_E \approx -1$

The two-phase Friedmann equation is

$$H^2(z) = H_0^2\Big[\Omega_m(1+z)^3 + \Omega_E(1+z)^{3(1+w_E)}\Big]$$

with $\Omega_E = 1-\Omega_m$. For $w_E = -1$ (PM prediction: ground-state medium has constant density) this is **mathematically identical to ΛCDM**.

The Origin of $\beta = 0.8$

Define the instantaneous power-law index

$$\beta_{\text{eff}}(z) = \frac{d\ln H}{d\ln(1+z)} = \frac{3}{2}\,\frac{\Omega_m(1+z)^3 + (1+w_E)\,\Omega_E(1+z)^{3(1+w_E)}}{E^2(z)}.$$

Key limits (Planck 2018, $w_E = -1$):

Limit	β_{eff}	Interpretation
$z \to \infty$	3/2	Pure matter domination
$z = 1$	1.18	Transition epoch
$z = 0$	$(3/2)\,\Omega_m \approx 0.47$	Energy-phase domination
$\langle\beta\rangle_{D_L}$, z=0.1–1.5	≈ 1.0	Instantaneous mean
Best-fit const. β to D_L	≈ 0.80	**PM calibration**

Conclusion: $\beta = 0.8$ is NOT a PM-specific parameter. It is the best-fit single power-law (constant β) to $D_L^{\Lambda\text{CDM}}$ over $z = 0.1$–1.5. The fit lands at $\beta \approx 0.8$ rather than 1.0 because errors partially cancel between regions where $\beta_{\text{eff}} < 0.8$ (low z) and $\beta_{\text{eff}} > 0.8$ (high z). Either way, the calibration was performed against ΛCDM data — the energy phase was implicitly assumed all along.

Implications

1. **PM cosmology requires dark energy.** Without an energy-phase component ($\Omega_E = 0$, single β), PM-Drag with $\Omega_{\text{eff}} = 1$ has no valid matter-dominated era (ODE failure; falsified).
2. **The two-phase model ($w_E = -1$) is ΛCDM.** All observables — luminosity distance, growth factor, $f\sigma_8$ — are numerically identical. This removes the previous free parameter Ω_{eff} and replaces it with a physically derived $\Omega_E = 1 - \Omega_m = 0.685$.
3. **Compact objects and cosmology share one phase transition.** The same $\varphi_{\text{crit}} = 1$ that caps neutron-star masses at $\approx 14\,M_\odot$ drives cosmic acceleration.
4. **The cosmological constant problem in PM language.** The energy-phase density at compact-object scales ($\rho_E \sim e\,\rho_{\text{nuc}} \approx 6\times10^{17}\,\text{kg/m}^3$) is $\sim 10^{44}\times$ larger than the observed cosmological value ($\rho_\Lambda \approx 6.5\times10^{-27}\,\text{kg/m}^3$). PM names the problem but does not solve it.
5. **Distinguishable prediction.** A slow matter→energy phase conversion at rate α gives $w_E > -1$ (quintessence-like), differing from the pure cosmological constant. Near-future surveys (DESI, Euclid) can measure w_E to ±0.05; PM predicts w_E very close to -1.

Implementation: `src/pushing_medium/cosmology.py` (`pm_twophase_hubble`, `pm_effective_beta`, `pm_energy_phase_fraction`, `pm_twophase_growth_factor_rate`, `pm_twophase_fsigma8`), `scripts/pm_twophase_cosmology.py`, `tests/test_cosmology_growth.py` (47 tests total).

Symbol	Value	Physical meaning
c	2.998×10^8 m/s	Speed of light
G	$6.674 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$	Newton's constant
μ_G	$1.486 \times 10^{-27}\ \mathrm{m^2\,kg^{-1}}$	Grav. index coeff. ($= 2G/c^2$)
k_{TT}	1.0 (exact)	GW power normalisation
k_{Fizeau}	1.0 (exact)	Moving-lens drag
A_M	$G/c^2 \approx 7.43 \times 10^{-28}\ \mathrm{m^3\,kg^{-1}\,s^{-1}}$	Mass-current coupling
A_q	$< 3 \times 10^{-10}\ \mathrm{m^3\,C^{-1}\,s^{-1}}$	Charge-current coupling
κ_B	≈ 0 (ruled out)	Magnetic potential coupling
ρ_{nuc}	$2.3 \times 10^{17}\ \mathrm{kg\,m^{-3}}$	Nuclear saturation density
Unconstrained: μ_E, K_E, B_q, K_{plasma}, α_{plasma}, c_ϕ, c_u ($\approx c$)		

Document Map

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